

# 01 Execution Evidence Report

Step-by-step build evidence, screenshots, validation, decisions, and ownership disclosure.

|                |                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository     | <a href="https://github.com/aalanani2000/al-rouf-ai-integration-assessment">https://github.com/aalanani2000/al-rouf-ai-integration-assessment</a> |
| Default branch | main                                                                                                                                              |
| Latest commit  | 7dcd3eadb125137db59fb128b337ba47cc1d1dbb                                                                                                          |
| Prepared date  | 2026-06-24                                                                                                                                        |
| Assessment     | AI Integration Engineer task pack                                                                                                                 |

## Assessment Requirements Covered

- Runnable GitHub repository with setup documentation, task folders, screenshots, and report deliverables.
- RFQ to CRM automation using n8n as an equivalent orchestration approach.
- FastAPI quotation microservice with tests, Docker packaging, and OpenAPI documentation.
- Bilingual RAG workflow over sample English and Arabic documents with citations and refusal behavior.
- Explicit notes on error handling, maintainability, security hygiene, trade-offs, validation, and AI assistance.

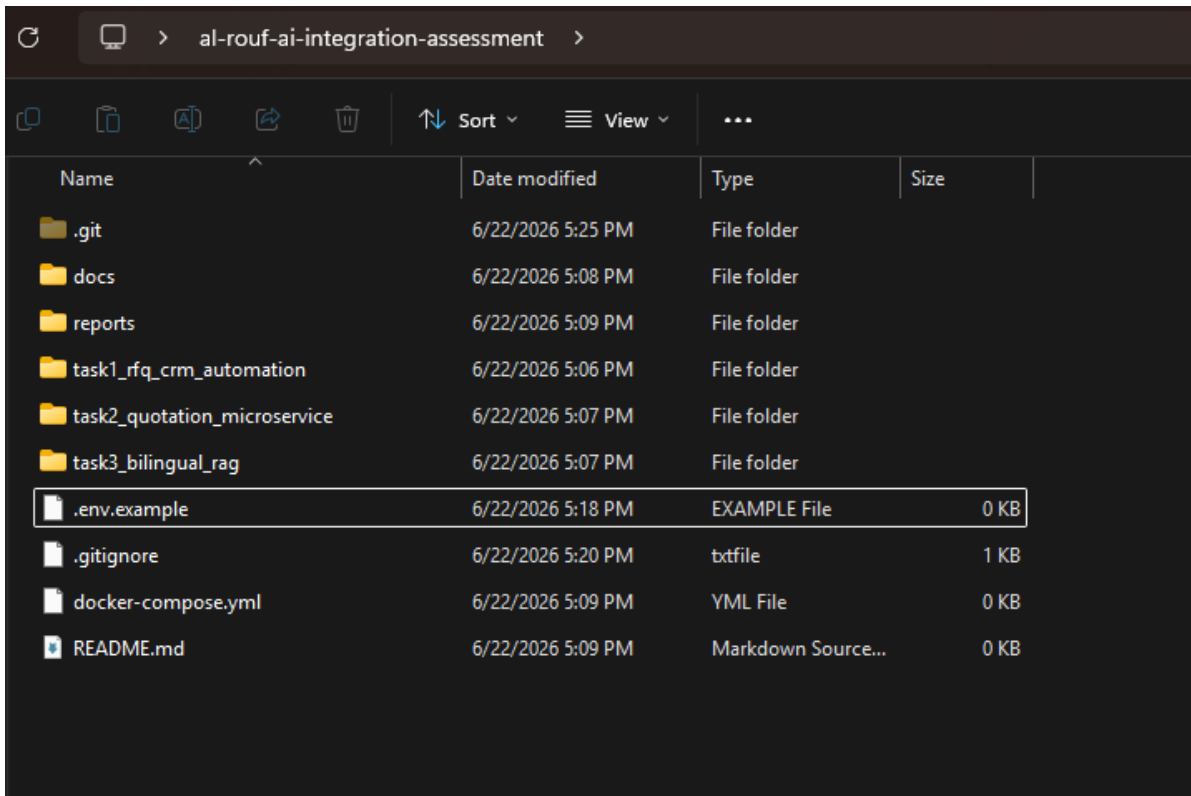
## Work Evidence By Task

| Task   | Implementation                                                                          | Validation                                                                              | Status                                                   |
|--------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|
| Task 1 | n8n RFQ webhook, OpenAI extraction, mock CRM, bilingual reply, alert log.               | Successful n8n executions, persisted JSON records, screenshot evidence.                 | Complete; attachment bytes scoped as future enhancement. |
| Task 2 | FastAPI quotation API with catalog, pricing tiers, SQLite, Docker, OpenAPI.             | Automated pytest suite and OpenAPI screenshot evidence.                                 | Complete.                                                |
| Task 3 | FastAPI bilingual RAG service over English/Arabic documents with citations and refusal. | Unit tests, integration test path, Arabic question evidence, error-handling screenshot. | Complete.                                                |

## Task 1 Evidence: RFQ To CRM Automation

1. Created a Docker-based n8n environment with a named n8n data volume and a bind-mounted local data folder.
2. Built a webhook intake endpoint for RFQ-style JSON payloads.

3. Configured an OpenAI node to extract urgency, key requirements, and a concise summary from unstructured RFQ text.
  4. Merged structured webhook fields with extracted intent and generated CRM metadata.
  5. Persisted records into a mock CRM JSON file and wrote sales alerts with bilingual reply drafts.
  6. Debugged n8n sandbox limitations, missing-file behavior, and node-reference mismatches during testing.
- Key trade-off: webhook simulation was used instead of email/IMAP to keep the workflow reproducible without OAuth setup.
  - Key limitation: attachment filenames are captured as metadata; real file-byte archiving is documented as a production enhancement.
  - Validation: multiple full workflow executions produced successful CRM and alert outputs.



| Name                         | Date modified     | Type               | Size |
|------------------------------|-------------------|--------------------|------|
| .git                         | 6/22/2026 5:25 PM | File folder        |      |
| docs                         | 6/22/2026 5:08 PM | File folder        |      |
| reports                      | 6/22/2026 5:09 PM | File folder        |      |
| task1_rfq_crm_automation     | 6/22/2026 5:06 PM | File folder        |      |
| task2_quotation_microservice | 6/22/2026 5:07 PM | File folder        |      |
| task3_bilingual_rag          | 6/22/2026 5:07 PM | File folder        |      |
| .env.example                 | 6/22/2026 5:18 PM | EXAMPLE File       | 0 KB |
| .gitignore                   | 6/22/2026 5:20 PM | txtfile            | 1 KB |
| docker-compose.yml           | 6/22/2026 5:09 PM | YML File           | 0 KB |
| README.md                    | 6/22/2026 5:09 PM | Markdown Source... | 0 KB |

Task 1 evidence: repository structure.png

```
PS C:\Users\idhoo\Desktop\al-rouf-ai-integration-assessment> docker inspect n8n --format '{{.Mounts}}'
[{"volume": "n8n_data", "n8n_data": "/home/node/.n8n", "local": "rw", "true": true}, {"bind": "C:\\Users\\idhoo\\Desktop\\al-rouf-ai-integration-assessment\\data", "data": "/data", "rw": true, "rprivate": true}]
```

Task 1 evidence: proof of the volumes being correctly attached.png

Webhook

Parameters Settings Docs

Webhook URLs

Test URL Production URL

POST http://localhost:5678/webhook-test/rfq-intake

HTTP Method

POST

Path

/rfq-intake

Authentication

None

Respond

When Last Node Finishes

Response Data

First Entry JSON

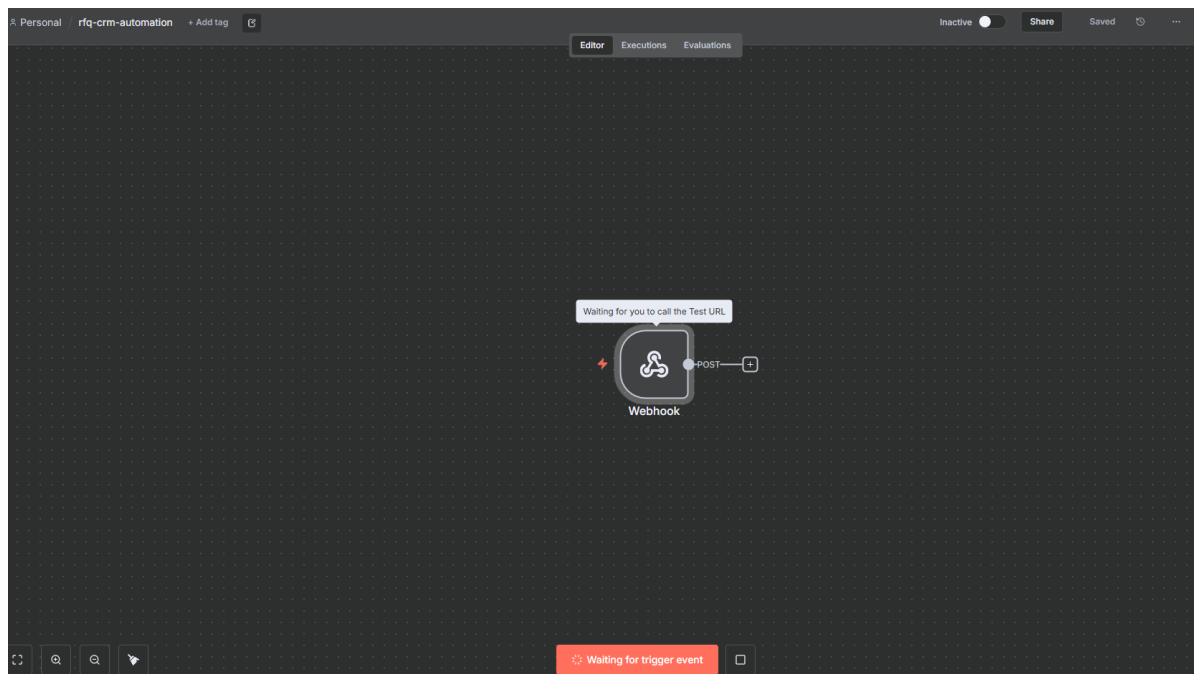
Options

No properties

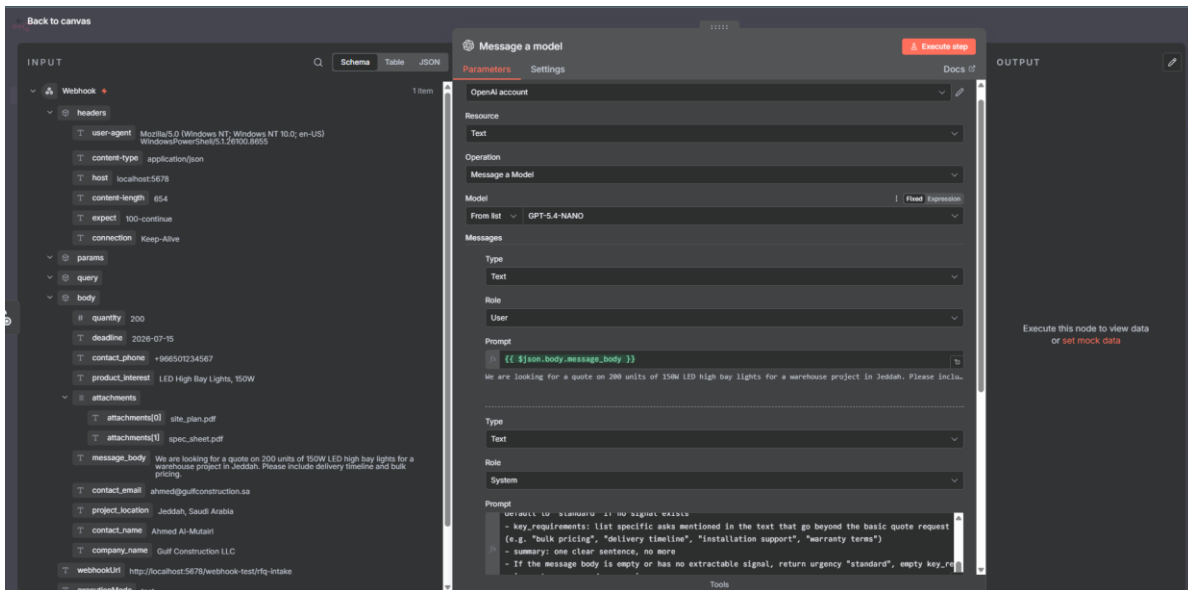
Add option

Listen for test event

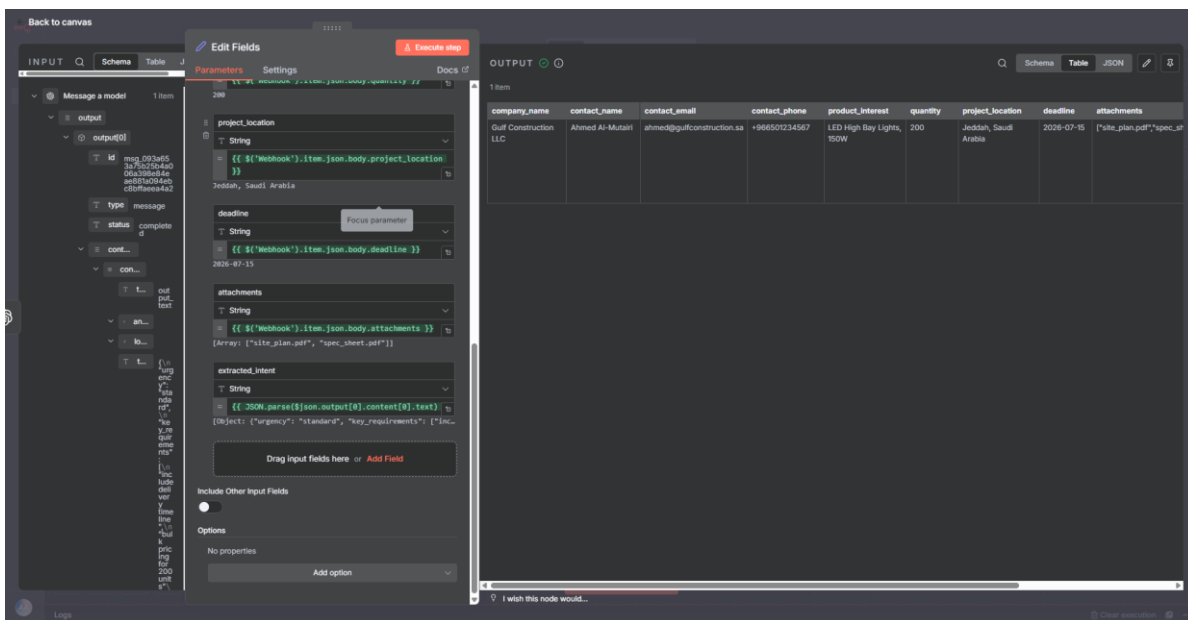
Task 1 evidence: Webhook node.png



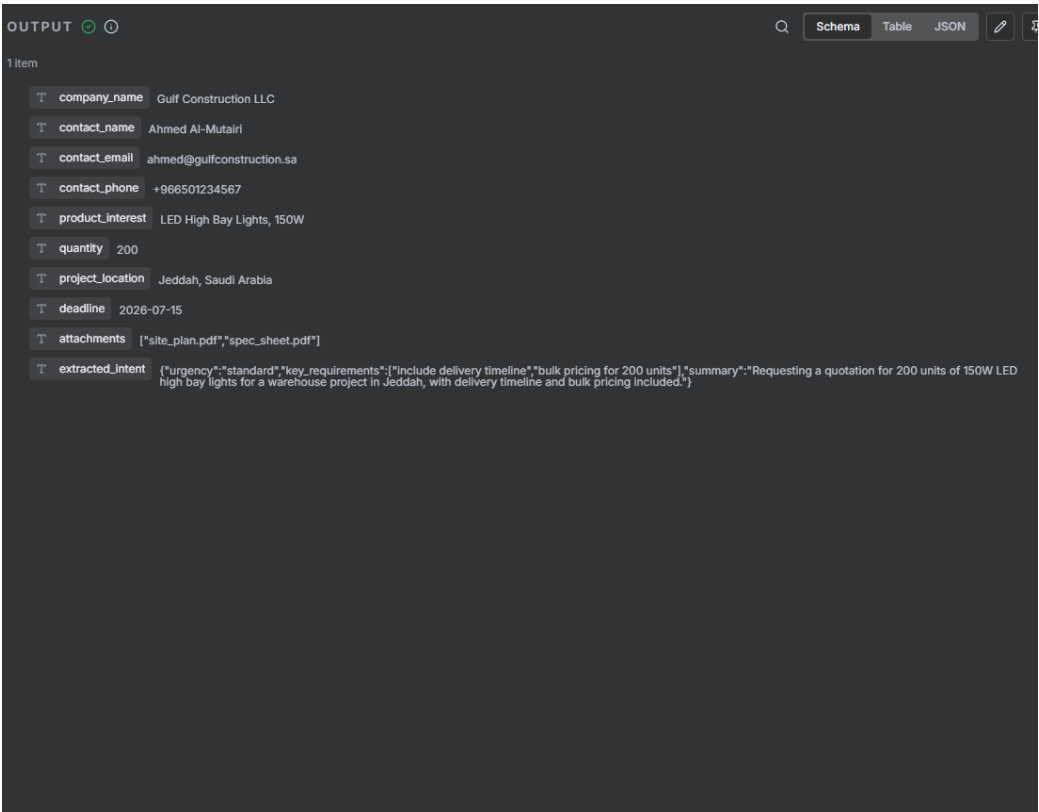
Task 1 evidence: n8n canva.png



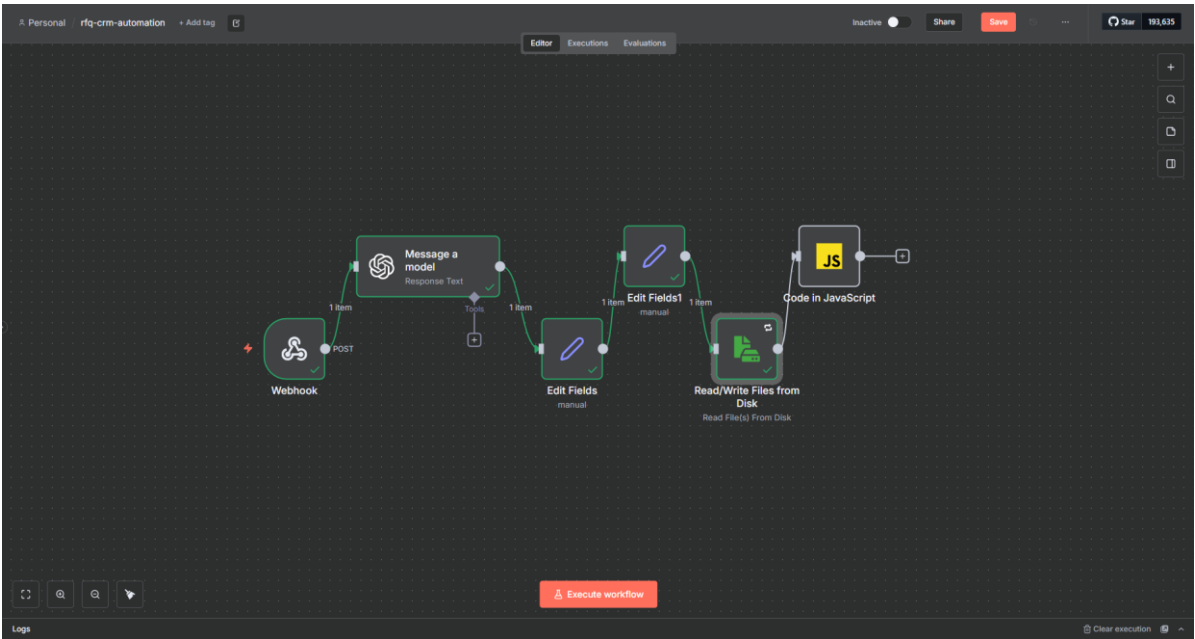
Task 1 evidence: open ai node.png



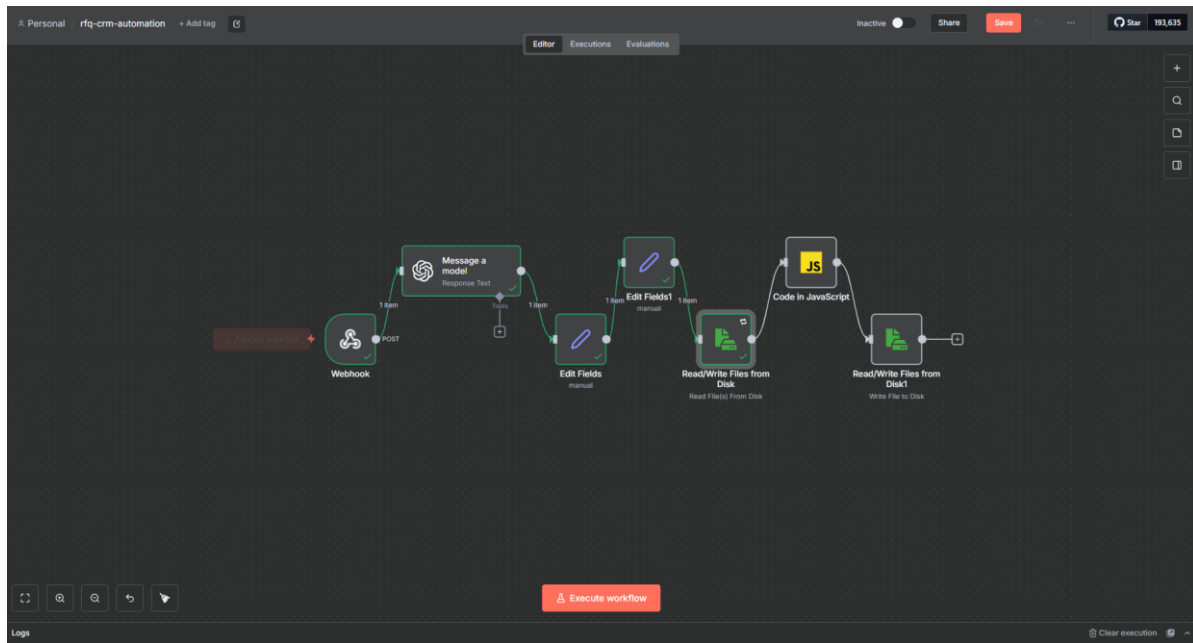
Task 1 evidence: set field.png



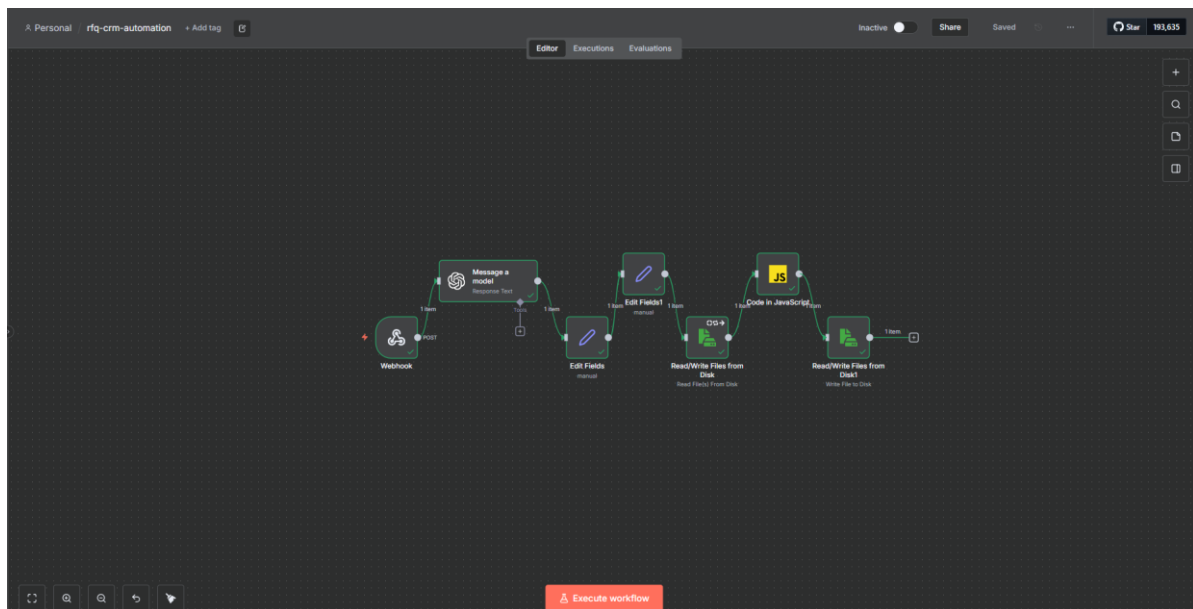
Task 1 evidence: set field output.png



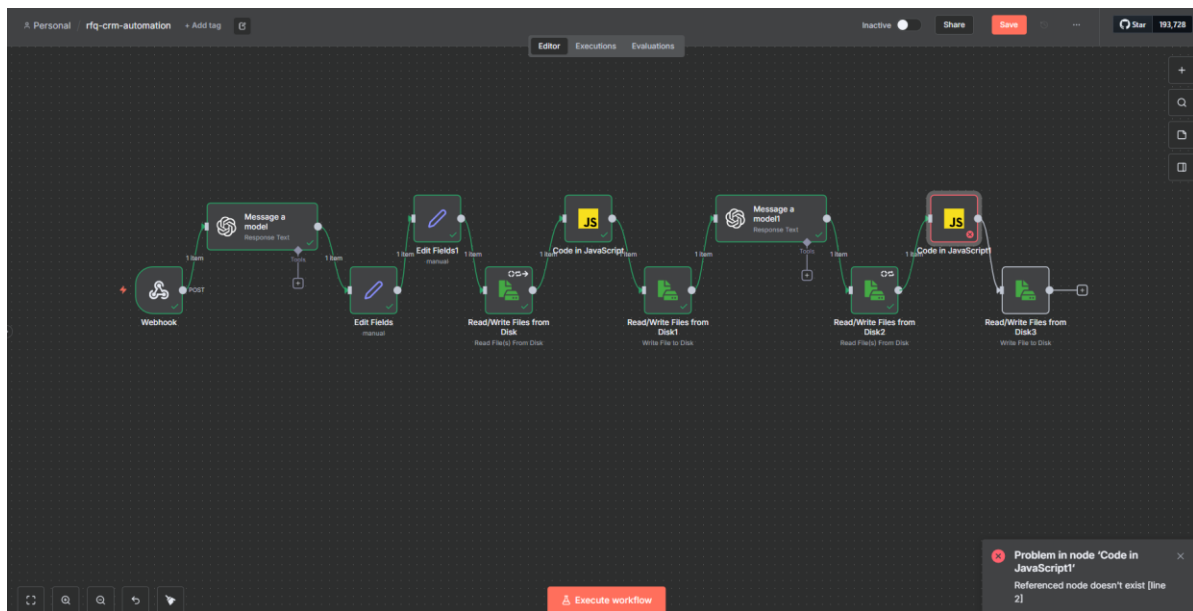
Task 1 evidence: workflow.png



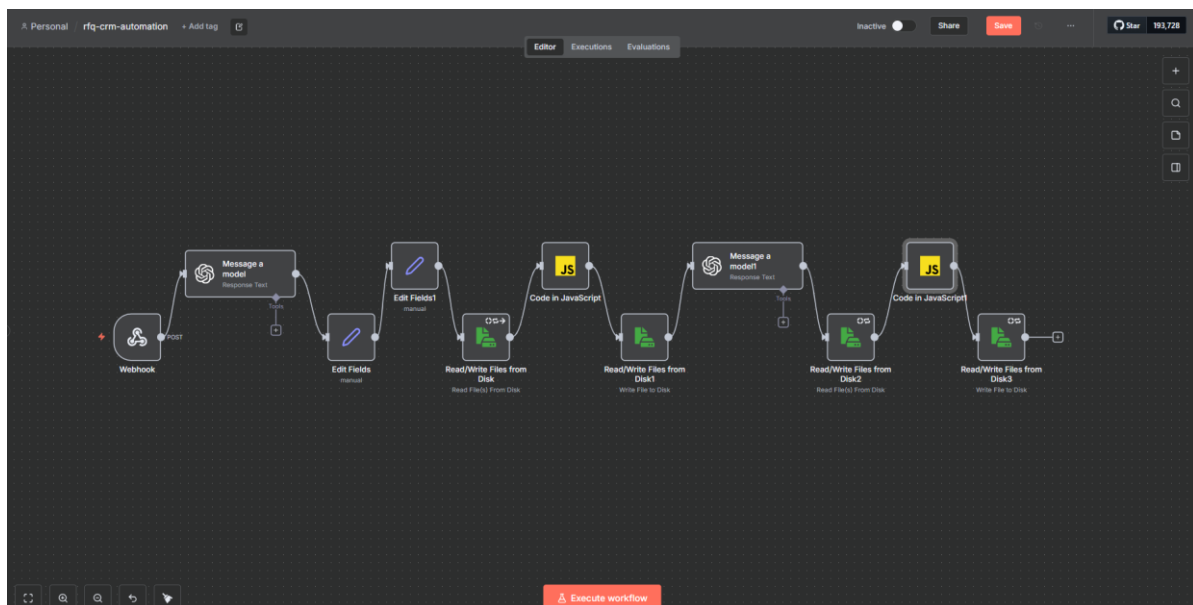
Task 1 evidence: workflow2.png



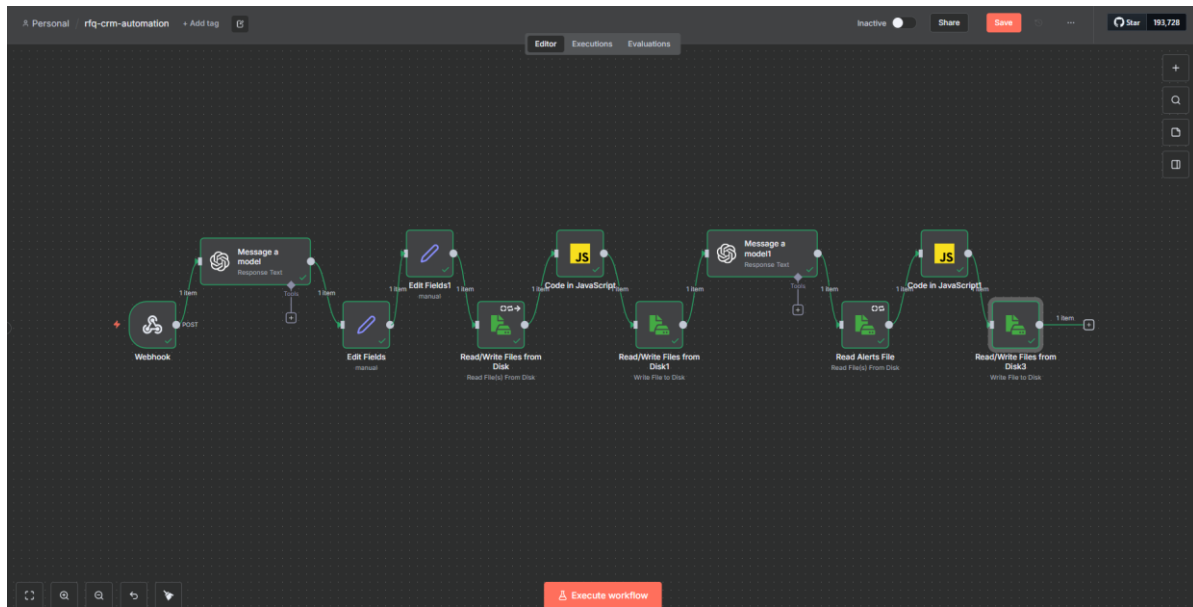
Task 1 evidence: workflow succes.png



Task 1 evidence: javascript code 1 error .png



Task 1 evidence: workflow progress .png

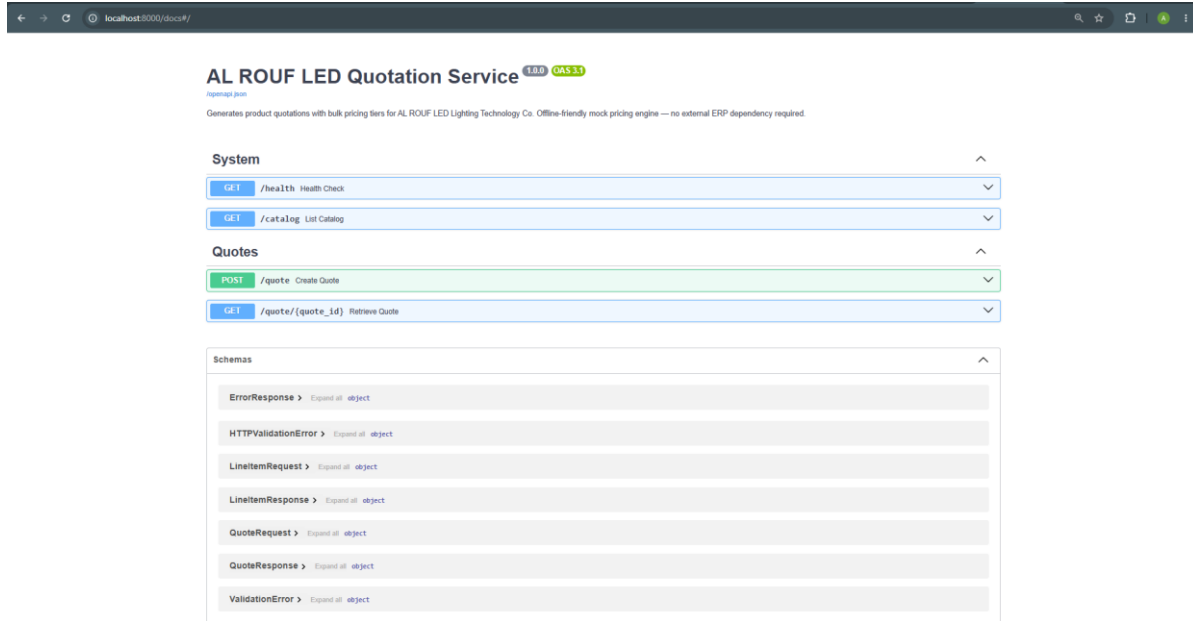


Task 1 evidence: workflow progress 1 .png

## Task 2 Evidence: Quotation Microservice

- Built a FastAPI service with `/health`, `/catalog`, `POST /quote`, and `GET /quote/{quote\_id}` endpoints.
- Implemented product catalog and bulk-discount pricing logic with offline-friendly mocked data.
- Persisted quotes in SQLite and packaged the service with Docker.
- Added automated tests for health, catalog, quote creation, discount tiers, validation failures, and quote retrieval.





Task 2 evidence: OpenAPI documentation page.

## Task 3 Evidence: Bilingual RAG Knowledge Workflow

- Built a FastAPI service that loads 4 sample documents in English and Arabic.
- Embedded document chunks with OpenAI embeddings and cached them locally.
- Retrieved relevant chunks using cosine similarity and refused out-of-scope questions before generation.
- Returned same-language answers with citations, latency, and estimated cost fields.
- Tested Arabic handling, retrieval behavior, refusal logic, and optional OpenAI-backed integration paths.

```
PS C:\Users\idhoo> $body = @{ question = "اعراض لـ داء ابيض بلع نامضلة قدم به ام" } | ConvertTo-Json
PS C:\Users\idhoo> Invoke-RestMethod -Uri "http://localhost:8001/ask" -Method Post -Body $body -ContentType "application/json"

question      : ?? ?? ??? ?????? ??? ?????? ?????????
answer        : I don't have enough information in the supported documents to answer this
               question confidently.
cited_sources : {}
confidence    : low
refused       : True
retrieved_matches : {@{source_doc=doc3_shipping_terms_en.md; similarity_score=0.132},
                   @{{source_doc=doc1_product_spec_high_bay_en.md; similarity_score=0.1011},
                   @{{source_doc=doc1_product_spec_high_bay_en.md; similarity_score=0.0982}}
latency_seconds : 0.0
estimated_cost_usd : 0.0
```

Task 3 evidence: model communication issue captured during testing.

Additional Task 3 evidence is stored at docs/screenshots/Task 3/testing the RAG on arabic question.pdf.

## Validation Summary

- Task 1 was validated through end-to-end n8n executions and persisted mock output files.
- Task 2 includes a pytest suite documented in its README; OpenAPI evidence confirms the interactive API surface.
- Task 3 includes offline tests for document loading, chunking, search, and refusal behavior, plus integration tests gated by `OPENAI\_API\_KEY`.
- Docker packaging is included for all runtime services that require containers.

## Ownership And AI Assistance Disclosure

AI assistance was used as a technical mentor and documentation partner for architecture discussion, starter logic, debugging support, and final writing polish. The candidate personally completed the hands-on implementation work: n8n node setup, Docker execution, API implementation, test execution, screenshot collection, environment configuration, validation, and final engineering decisions.